

PGR210 – NLP part

Kristiania University College

By Huamin Ren

Huamin.ren@kristiania.no

Lec 1

Huamin

Outline



Introduction to this course



Introduction to Natural Language Processing (NLP)



Environment setup: Pycharm, Anaconda, Jupyter notebook



Popular NLP libraries



NLP Pipeline

1.1. Course Schedule

Lectures			Practices
Part I: Text preprocessing			
NLP-W1	Lecture 1	1.1 Introduction to NLP	P1
		1.2 Popular NLP libraries	
		1.3 Use cases and applications	
		1.4 NLP Pipeline	
	Lecture 2	2.1 Text processing: removing HTML tags, text tokenization, removing special characters, case conversions, stemming, normalization, removing stop words	P2
		2.2 Text visualization and basic analysis	
		2.3 Text processing I	
NLP-W2	Lecture 3	3.1 Text processing II	P3
	Lecture 4	4.1 Text representation: Introduction to text representation, BoW	P4
		4.2 Text representation II: TF-IDF representation	
Part II: Text representation			
NLP-W3	Lecture 5	5.1 Word embeddings	P5
		5.2 Visualization	
	Lecture 6	6.1 K-means	P6
		6.2 Bring all processing together with representation and K-means	
NLP-W4	Lecture 7	7.1 Looking into semantics: text classifier	P7
		7.2 Topic modelling	
	Lecture 8	8.1 LSA, SVD and LDA 8.2 An application	P7
Part III: Advanced topics			
NLP-W5	Lecture 9	9.1 Learning the history of neural networks	P9

Lecture/Exercise 1 & 2 (week 1): Introduction to NLP, popular NLP libraries, use cases and applications, NLP pipeline; Text visualization and basic analysis, text processing I

Lecture/Exercise 3 & 4 (week 2): Text processing II, introduction to text representation: BoW, TF-IDF representation

Lecture/Exercise 5 & 6 (week 3): word embeddings and visualization; K-means clustering

Lecture/Exercise 7 & 8 (week 4): Looking into semantic, topic modelling, LSA, SVD and LDiA

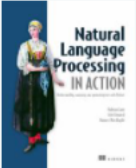
Lecture/Exercise 9 & 10 (week 5): Understanding neurals and backpropagation, and then build up a basic network in Keras

Lecture/Exercise 11 & 12 (week 6): A mini project on NLP application, and recent progress and future trends

1.2 Textbook and further reading materials

- Leganto: <https://bibsys-xm.alma.exlibrisgroup.com/leganto/readinglist/lists/8479045600002296?auth=SAML>

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BOOK

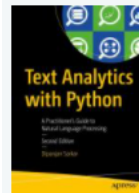
Natural language processing in action : understanding, analyzing, and generating text with Python ✓

Title: Natural language processing in action : understanding, analyzing, and generating text with Python
Author: Hobson, Lane
ISBN: 9781617294631; 1617294632
Total Pages: xxix, 512 sider:
Publication Date: [2019]
Publisher: Manning Publications Co
Place of Publication: Shelter Island, NY :

[More details](#) ▾

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BOOK

Text Analytics with Python [electronic resource] : A Practitioner's Guide to Natural Language Processing ✓

Title: Text Analytics with Python [electronic resource] : A Practitioner's Guide to Natural Language Processing

Author: Sarkar, Dipanjan.

ISBN: 1-4842-4354-4

Edition: 2nd ed. 2019.

Total Pages: 1 online resource (688 pages)

Publication Date: 2019.

Publisher: Apress ; Imprint Apress

Place of Publication: Berkeley, CA :

[More details ▾](#)

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Further reading

- **Natural Language Processing with Python**

By [Steven Bird](#), [Ewan Klein](#), [Edward Loper](#)

<https://learning.oreilly.com/library/view/natural-language-processing/9780596803346/>

- **Practical Natural Language Processing**

By Sowmya Vajjala and etc

A comprehensive guide to building real-world NLP systems

1.3 High-level plan: mandatory HW and exam

- <https://www.kristiania.no/studieportal/school-of-economics-innovation-and-technology/bachelorniva/pgr210/machine-learning-and-natural-language-processing/>
- Exam

1.4 Course requirement

- Actively participate: discussion, presentation, group work
- Written home examination *in group*
- Giving feedback to TAs is encouraged

Mykhailo, Hansim

- Prior programming experience assumed; basic statistics and data analysis skills required

1.5 Lectures & Practices

- 10:15-11:00: Lecture session
- 11:15-12:00: Lecture session
- 12:15-14:00: Practice

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NLP Pipeline

2.1. What is NLP

- What is Natural Language?

To understand text analytics and natural language processing , we need to understand what makes a language “natural.” In simple terms, a natural language is one developed and evolved by humans through ***natural use*** and ***communication*** , rather than constructed and created artificially, like a computer programming language.

2.1. What is NLP

- Definition: Natural language processing is an area of research in computer science and artificial intelligence (AI) concerned with processing natural languages such as English or Mandarin.
- This processing generally involves:
 - translating natural language into data (numbers) that a computer can use to learn about the world.
 - This understanding of the world is sometimes used to generate natural language text that reflects that understanding.

2.2 Natural Language Processing System

- Often referred to as a *pipeline* because it usually involves several stages of processing where natural language flows in one end and the processed output flows out the other

2.3 What's so special about human (natural) language?

- Human language is a system specifically constructed to convey meaning, and is not produced by a physical manifestation of any kind. In that way, it is very different from vision or any other machine learning task.

the symbols of language can be encoded in several modalities : voice, gesture, writing, etc

2.4 Magic

- Natural languages can't be directly translated into a precise set of mathematical operations
- But they do contain information and instructions that can be extracted
- Those pieces of information and instruction can be stored, indexed, searched, or immediately acted upon

2.4.1 Discussion #1

- Is natural language processing more difficult than a programming language? And why?

2.4.1 Discussion #2

Powerful assumptions:

‘good morning’:

- assume that you have some knowledge about what makes up a morning, including not only that mornings come before noons and afternoons and evenings but also after midnights;
- Assume you know they can represent times of day as well as general experiences of a period of time;
- assumed you know that “good morning” is a common greeting that doesn’t contain much information at all about the morning.

Discussions in pairs:

One: ask a question

The other: answer the question

In together: analyze which assumptions are required to answer the question

Practice in three questions, from easy-medium-difficult levels

2.4.2 The powerful ‘processor’

- The “processor” has access to a lifetime of common sense knowledge about the world
 - ✓ Ontologies
 - ✓ Knowledge bases
 - ✓ Common sense knowledge

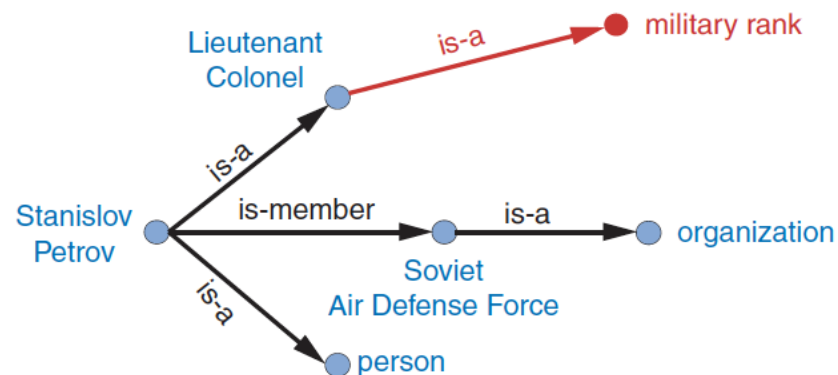
2.4.2 The powerful ‘processor’

- The “processor” has access to a lifetime of common sense knowledge about the world

✓ **Ontologies**

✓ Knowledge bases

✓ Common sense knowledge



Example:

In 1983, Stanislav Petrov, a lieutenant colonel of the Soviet Air Defense Forces, saved the world from nuclear war.

(‘Stanislav Petrov’, ‘is-a’, ‘lieutenant colonel’)

a graph of triplets in the form of (subject, relation, object)

https://en.wikipedia.org/wiki/Stanislav_Petrov

2.3.2 The powerful ‘processor’

- The “processor” has access to a lifetime of common sense knowledge about the world
 - ✓ Ontologies
 - ✓ **Knowledge bases**
 - ✓ Common sense knowledge

Knowledge graph (knowledge bases) would have to already contain facts about militaries and military ranks. It even include facts about the titles of people and how people relate to occupations (jobs).

Example:

In 1983, Stanislav Petrov, a lieutenant colonel of the Soviet Air Defense Forces, saved the world from nuclear war.



('Stanislav Petrov', 'is-a', 'lieutenant colonel')



a graph of triplets in the form of (subject, relation, object)

2.3.2 The powerful ‘processor’

- The “processor” has access to a lifetime of common sense knowledge about the world
 - ✓ Ontologies
 - ✓ Knowledge bases
 - ✓ **Common sense knowledge**

Discussion:

When do you think you start to acquire much of your common sense knowledge?

Interesting readings

- Representing Commonsense Knowledge

Nils J. Nilsson, in Artificial Intelligence: A New Synthesis, 1998

- Acquisition of Commonsense Knowledge

Erik T. Mueller, in Commonsense Reasoning (Second Edition), 2015

19.7 How big is human common sense?

How much commonsense knowledge do we need to acquire? How big is human commonsense knowledge? Landauer (1986) produced estimates of the size of an adult's long-term memory ranging from 0.5×10^9 to 3.4×10^9 bits using different methods:

1. Several estimates were based on the rate information is added to long-term memory—1.2 bits per second for text and 2.3 bits per second for pictures, from psychological experiments—and a linear accumulation over 70 years.
2. Another estimate took into account forgetting.
3. Another estimate was based on 346 bits to define each word in a dictionary containing 10^5 words, multiplied by 15 to take account of other domains of knowledge such as history, music, nature, personal life history, and motor skills.

Estimates ventured over the years are summarized in Table 19.2. For comparison, the Knext system has extracted 10^8 factoids (J. M. Gordon, Van Durme, & Schubert, 2010, p. 10).

18.1.2 Difficulties in Representing Commonsense Knowledge

What are some of the reasons that it has proved difficult to formalize commonsense knowledge? Probably one reason is its sheer bulk. Much expert knowledge can be compartmentalized in such a way that a few hundred or a few thousand facts are sufficient to build useful expert systems. How many will be needed by a system capable of general human-level intelligence? No one knows for sure. Doug Lenat, who is bravely leading an effort to build a large knowledge base of such facts, called CYC, thinks that between one and ten million will be needed [Guha & Lenat 1990, Lenat & Guha 1990, Lenat 1995]. In 1990, the CYC authors said:

Perhaps the hardest truth to face, one that AI has been trying to wriggle out of for 34 years, is that there is probably no elegant, effortless way to obtain this immense knowledge base. Rather, the bulk of the effort must (at least initially) be manual entry of assertion after assertion. [Guha & Lenat 1990, p. 33]

Another difficulty is that commonsense knowledge seems not to have well-defined frontiers that enable us to get a grip on parts of it independently of the other parts. Conceptualizations of the commonsense world would probably involve many entities, functions, and relations that would crop up diffusely throughout the conceptualization. Thus, as we attempt to develop a conceptualization, we wouldn't know whether we had “gotten it right” until we had nearly finished the entire, exceedingly large job.

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3.1 Anaconda3

- Python 3 has a lot of performance and expressiveness features that are handy for NLP.
- The easiest way to install Python 3 on almost any system is to install Anaconda3 (<https://www.anaconda.com>).

This has the added benefit of giving you *a package and environment manager* that can install a lot of problematic packages (such as `matplotlib`) on a wide range of problematic OSes (like Windows).

<https://docs.anaconda.com/anaconda/install/mac-os/>

Verify your installation!

3.2 IDE

- What you need is only a good text editor to round out your integrated development environment (IDE).
- Pycharm is used during the course, but other options are open (your decision): <https://www.jetbrains.com/help/pycharm/installation-guide.html>
 - jupyter notebook

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NLP Pipeline

4.1 Natural Language Processing Frameworks

- There are dedicated frameworks and libraries for natural language processing and text analytics, which you can just install and start using, just like any other built-in module in the Python standard library.
- These frameworks have been built over a long period of time and are usually still in active development.

4.2 NLTK

- NLTK: <https://www.nltk.org/api/nltk.html>
- Install package in PyCharm
- Further reading: <https://www.nltk.org/book/>

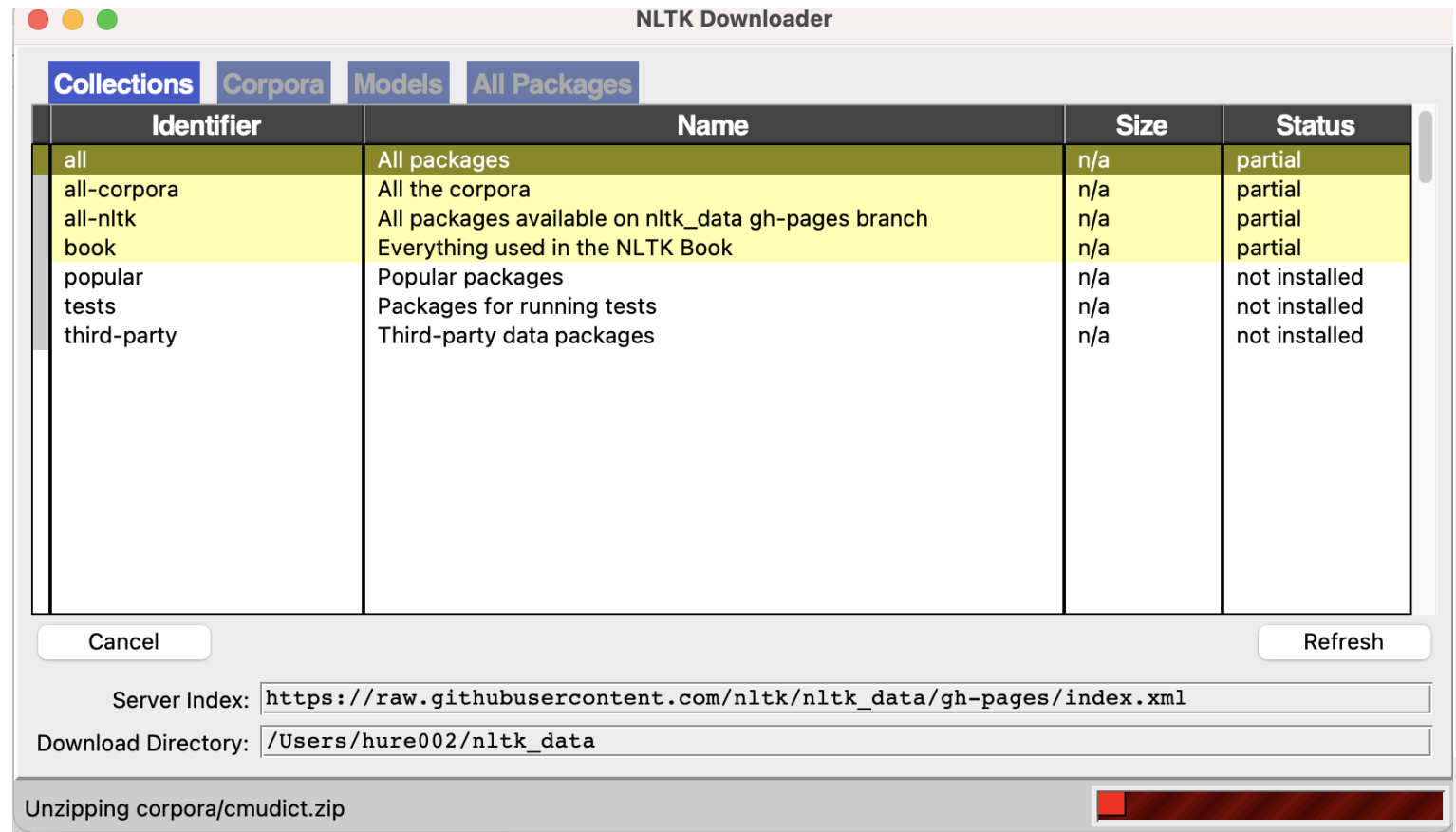
It is the *standard workhorse* of any NLP project in the industry.

- Using cmd to execute the code:

```
Last login: Thu Mar 18 08:31:49 on console
(base) hure002@ella006egmsno-C02DXCHHMD6N ~ % python --version
Python 3.8.5
(base) hure002@ella006egmsno-C02DXCHHMD6N ~ % python
Python 3.8.5 (default, Sep  4 2020, 02:22:02)
[Clang 10.0.0 ] :: Anaconda, Inc. on darwin
Type "help", "copyright", "credits" or "license" for more information.
[>>> import nltk
[>>> nltk.download()
showing info https://raw.githubusercontent.com/nltk/nltk_data/gh-pages/index.xml
```

- A corpora should be downloaded:

`nltk.download()`



- Information to show Stopwords Corpus:
<https://www.nltk.org/book/ch02.html>

4.3 GENSIM

- The gensim library has a rich set of capabilities for semantic analysis, including topic modeling and similarity analysis.
 - It contains a Python port of Google's very popular Word2Vec model, which is a neural network model implemented to learn distributed representations of words where similar words (semantic) occur close to each other.
- Popularly used in
 - semantic analysis
 - feature engineering

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2.4. Practical applications

- Natural language processing is everywhere.

Table 1.1 Categorized NLP applications

Search	Web	Documents	Autocomplete
Editing	Spelling	Grammar	Style
Dialog	Chatbot	Assistant	Scheduling
Writing	Index	Concordance	Table of contents
Email	Spam filter	Classification	Prioritization
Text mining	Summarization	Knowledge extraction	Medical diagnoses
Law	Legal inference	Precedent search	Subpoena classification
News	Event detection	Fact checking	Headline composition
Attribution	Plagiarism detection	Literary forensics	Style coaching
Sentiment analysis	Community morale monitoring	Product review triage	Customer care
Behavior prediction	Finance	Election forecasting	Marketing
Creative writing	Movie scripts	Poetry	Song lyrics

GROUP PRESENTATION:

Have you heard of all these applications?

Pick one application that sounds most unfamiliar to you, and seek more introduction information to prepare a group presentation.

2.4.1. Examples of tasks

Easy

- Spell checking
- Keyword search
- Finding synonyms

Medium

- Parsing information from websites, documents

Hard

- Machine translation
- Semantic analysis
- Question answering

2.4.2. Representative application on hard ones

- Machine translation

<http://demo.statmt.org/index.cgi>

Representative application on hard ones

- Semantic analysis

*Recent progress on Transformer a



ChatGPT4 does not currently exist as it is a hypothetical model, but I can explain how the transformer and attention models have contributed to the development of the previous versions of GPT models.



The transformer model is a type of neural network architecture that was introduced in the paper "Attention is All You Need" by Vaswani et al. (2017). It has revolutionized the field of NLP by providing an alternative to the traditional sequence-to-sequence models, such as the LSTM and GRU models. The transformer model is based on a self-attention mechanism that allows the model to focus on the most relevant parts of the input sequence at each step of the computation. This makes it more efficient and effective at capturing long-range dependencies in the input sequence.

The transformer architecture has been used in several language models, including GPT-1, GPT-2, and GPT-3, which are large-scale pre-trained language models developed by OpenAI. These models have achieved state-of-the-art performance on several NLP benchmarks, such as language modeling, text generation, and question answering.

In particular, the attention mechanism has enabled these models to capture complex linguistic structures and relationships between words and sentences, which has improved their ability to generate coherent and contextually appropriate responses in conversational settings. This has made them particularly useful for developing conversational AI

Representative application on hard ones

- Question answering

https://colab.research.google.com/github/spark-ming/albert-qa-demo/blob/master/Question_Answering_with_ALBERT.ipynb#scrollTo=rkivu8FOqp_8

Representative application on hard ones

- Text to speech

https://cloud.google.com/text-to-speech/?utm_source=google&utm_medium=cpc&utm_campaign=emea-no-all-en-dr-bkws-all-all-trial-e-gcp-1010042&utm_content=text-ad-none-any-DEV_c-CRE_495029614056-ADGP_Hybrid%20%7C%20BKWS%20-%20EXA%20%7C%20Txt%20~%20AI%20%26%20ML%20~%20Text-to-Speech%23v4-KWID_43700060416646650-aud-606988878614%3Akwd-477477965678-userloc_1011021&utm_term=KW_text%20to%20speech%20google-NET_g-PLAC_&gclid=Cj0KCQjw0caCBhCIARIsAGAfUMzk25tynK3sSblfA1rrvsThxcKopx30fjChRQgjH9s54EnN6jII5vcaAoGVEALw_wcB&gclsrc=aw.ds

A representative application

- 2018, Google, Wavenet:

<https://www.youtube.com/watch?v=JjK8apEishQ&t=175s>

<https://deepmind.com/blog/article/wavenet-generative-model-raw-audio>

- Clone voices:

https://www.youtube.com/watch?v=-O_hYhToKoA

Steps:

git clone <https://github.com/CorentinJ/Real-Time-Voice-Cloning.git>

Demos

- Fake voice text to speech using deep learning:

<https://www.youtube.com/watch?v=6bFN2YkN6bo>